Table 2a. Mali: Public Sector Debt Sustainability Framework, Baseline Scenario, 2018–38

(In percent of GDP, unless otherwise indicated

	Actual		Average 5/	Standard Deviation 5/	Projections									
	2016	2017			2018	2019	2020	2021	2022	2023	2018-23 Average	2028	2038	2024-38 Average
Public sector debt 1/	35.9	35.6			35.9	36.6	37.6	38.6	39.8	40.4		44.0	51.1	
of which: foreign-currency denominated	24.9	24.5			23.4	23.2	23.4	23.6	24.1	24.4		28.0	35.1	
Change in public sector debt	5.2	-0.3			0.3	0.7	1.0	1.0	1.2	0.6		0.9	0.5	
Identified debt-creating flows	2.5	-2.1			-0.1	0.5	0.5	0.4	0.5	0.4		0.3	-0.3	
Primary deficit	3.3	2.0	2.1	0.9	2.6	2.2	2.1	2.0	2.0	1.9	2.1	1.9	1.7	1.8
Revenue and grants	18.3	20.1			20.6	20.3	20.7	21.1	21.7	22.3		23.0	23.9	
of which: grants	1.6	1.6			1.2	1.6	1.7	1.8	1.8	1.8		1.4	0.6	
Primary (noninterest) expenditure	21.6	22.1			23.1	22.4	22.8	23.2	23.6	24.2		24.9	25.6	
Automatic debt dynamics	-0.8	-4.1			-2.6	-1.5	-1.5	-1.6	-1.5	-1.5		-1.6	-2.1	
Contribution from interest rate/growth differential	-1.4	-1.5			-1.7	-1.5	-1.4	-1.4	-1.4	-1.5		-1.1	-1.3	
of which: contribution from average real interest rate	0.3	0.3			0.0	0.2	0.3	0.3	0.3	0.3		0.8	0.9	
of which: contribution from real GDP growth	-1.7	-1.8			-1.7	-1.6	-1.6	-1.7	-1.7	-1.8		-1.9	-2.3	
Contribution from real exchange rate depreciation	0.6	-2.5			-0.9	0.0	-0.1	-0.2	0.0	-0.1		...	...	
Other identified debt-creating flows	0.0	-0.1			0.0	-0.2	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.2	0.1			0.2	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-0.2	-0.2			-0.2	-0.2	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	2.7	1.8			0.4	0.2	0.5	0.6	0.7	0.2		0.6	0.9	
Other Sustainability Indicators														
PV of public sector debt	...	26.8			27.6	28.3	29.1	29.9	30.7	31.1		32.8	36.1	
of which: foreign-currency denominated	...	15.8			15.1	14.9	14.9	14.8	15.0	15.1		16.7	20.1	
of which: external	...	15.8			15.1	14.9	14.9	14.8	15.0	15.1		16.7	20.1	
Gross financing need 2/	7.9	6.8			5.4	5.4	5.5	5.6	5.7	5.7		5.8	6.0	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	133.3			134.0	139.4	140.6	141.4	141.7	139.6		142.7	151.4	
PV of public sector debt-to-revenue ratio (in percent)	...	145.0			142.4	151.6	153.2	154.5	154.5	152.0		151.9	155.3	
of which: external 3/	...	85.3			77.8	80.0	78.2	76.7	75.6	73.8		77.6	86.3	
Debt service-to-revenue and grants ratio (in percent) 4/	9.6	10.7			9.1	10.6	11.0	11.4	11.3	11.2		10.9	11.9	
Debt service-to-revenue ratio (in percent) 4/	10.5	11.6			9.7	11.5	12.0	12.5	12.4	12.2		11.6	12.2	
Primary deficit that stabilizes the debt-to-GDP ratio	-2.0	2.4			2.2	1.5	1.1	1.0	0.8	1.3		1.0	1.2	
Key macroeconomic and fiscal assumptions														
Real GDP growth (in percent)	5.8	5.3	4.4	2.3	5.0	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7
Average nominal interest rate on forex debt (in percent)	1.5	1.5	1.5	0.2	1.0	1.1	1.2	1.2	1.3	1.3	1.2	1.4	1.5	1.5
Average real interest rate on domestic debt (in percent)	3.3	3.5	1.8	3.5	3.4	3.7	3.4	3.2	3.0	2.9	3.3	3.0	2.8	2.9
Real exchange rate depreciation (in percent, + indicates depreciation)	2.9	-10.7	0.0	8.5	-3.9	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	1.5	1.6	4.1	3.4	1.4	1.4	1.8	2.0	2.0	2.2	1.8	2.0	2.2	2.2
Growth of real primary spending (deflated by GDP deflator, in percent)	12.1	8.2	3.2	5.2	9.8	1.5	6.3	6.4	6.8	7.2	6.3	4.9	4.8	5.1
Grant element of new external borrowing (in percent)	...	...	...	...	48.0	48.0	48.0	48.0	48.0	48.0	48.0	48.0	48.0	48.0

Sources: Country authorities; and staff estimates and projections.
1/ Gross debt of central government

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability. The historical average for the primary deficit, however, excludes 2006 (the year of MDRI debt relief and hence an unusually large primary surplus).

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Table 2b. Mali: Sensitivity Analysis for Key Indicators of Public Debt, 2018–38

	Projections							
	2018	2019	2020	2021	2022	2023	2028	2038
PV of Debt-to-GDP Ratio								
Baseline	28	28	29	30	31	31	33	36
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	28	28	29	30	31	32	34	40
A2. Primary balance is unchanged from 2018	28	29	30	31	32	33	36	42
A3. Permanently lower GDP growth 1/	28	28	30	31	32	33	37	51
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-2020	28	29	31	33	34	35	39	47
B2. Primary balance is at historical average minus one standard deviations in 2019-2020	28	29	30	31	32	32	34	37
B3. Combination of B1-B2 using one half standard deviation shocks	28	29	30	32	33	33	37	42
B4. One-time 30 percent real depreciation in 2019	28	34	34	34	34	34	33	33
B5. 10 percent of GDP increase in other debt-creating flows in 2019	28	34	35	35	36	36	37	39
PV of Debt-to-Revenue Ratio 2/								
Baseline	134	139	141	141	142	140	143	151
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	134	140	141	143	144	142	150	168
A2. Primary balance is unchanged from 2018	134	141	143	145	147	146	156	177
A3. Permanently lower GDP growth 1/	134	140	143	145	147	146	162	212
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-2020	134	144	151	154	157	157	171	197
B2. Primary balance is at historical average minus one standard deviations in 2019-2020	134	142	146	146	146	144	146	153
B3. Combination of B1-B2 using one half standard deviation shocks	134	142	147	149	150	149	159	177
B4. One-time 30 percent real depreciation in 2019	134	168	165	161	158	152	144	137
B5. 10 percent of GDP increase in other debt-creating flows in 2019	134	167	167	166	165	162	161	162
Debt Service-to-Revenue Ratio 2/								
Baseline	9	11	11	11	11	11	11	12
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	9	11	11	12	11	11	11	13
A2. Primary balance is unchanged from 2018	9	11	11	12	11	11	11	13
A3. Permanently lower GDP growth 1/	9	11	11	12	12	12	12	15
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-2020	9	11	12	12	12	12	12	14
B2. Primary balance is at historical average minus one standard deviations in 2019-2020	9	11	11	12	11	11	11	12
B3. Combination of B1-B2 using one half standard deviation shocks	9	11	11	12	12	12	11	13
B4. One-time 30 percent real depreciation in 2019	9	12	13	14	13	13	13	15
B5. 10 percent of GDP increase in other debt-creating flows in 2019	9	11	12	12	12	12	12	13

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.

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